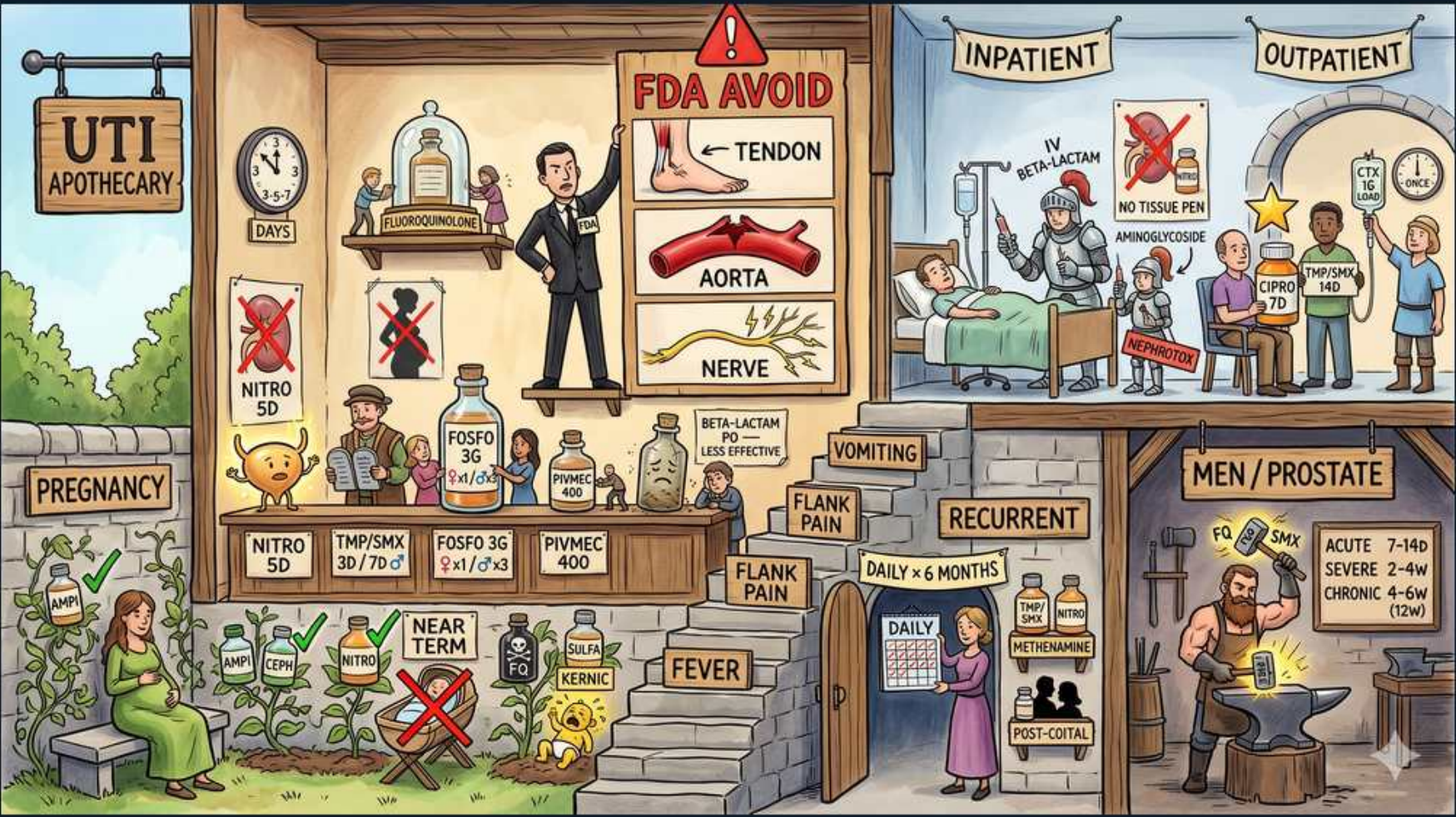


UTI — Antibiotic Treatment by Setting & Host (Apothecary)



1. Nitrofurantoin 5 d

WHAT YOU SEE

'NITRO 5D' bottle on apothecary shelf

WHAT IT ENCODES

Nitrofurantoin (100 mg BID x5 days) is a first-line agent for acute uncomplicated cystitis. It concentrates in the URINE only, so it works for cystitis but NOT for pyelonephritis or any systemic/tissue infection. Avoid when CrCl <30 mL/min (older cutoffs said <60) and avoid at term and in G6PD deficiency (hemolysis).

BOARD TRAP

Nitrofurantoin does NOT achieve therapeutic tissue/blood levels — never use it for pyelonephritis or prostatitis. It also fails in renal impairment (low urinary concentration) and is contraindicated near term.

2. TMP-SMX 3 d

WHAT YOU SEE

'TMP/SMX 3D / DS' bottle

WHAT IT ENCODES

Trimethoprim-sulfamethoxazole (1 DS tablet BID x3 days) is first-line for uncomplicated cystitis WHERE local E. coli resistance is <20%. Unlike nitrofurantoin it does penetrate tissue, so it also treats pyelonephritis (longer course). Avoid in late pregnancy (kernicterus) and watch for hyperkalemia and sulfa allergy.

BOARD TRAP

If local resistance exceeds ~20%, TMP-SMX is no longer reliable empiric therapy — and avoid it near term. It's a tissue-penetrating agent, so unlike nitrofurantoin it CAN treat pyelonephritis.

3. Fosfomycin 3 g single dose

WHAT YOU SEE

'FOSFO 3G x1' single-dose sachet

WHAT IT ENCODES

Fosfomycin (3 g single oral dose) is first-line for uncomplicated cystitis, valued for single-dose convenience and activity against many multidrug-resistant organisms including ESBL E. coli and VRE. Like nitrofurantoin it is a urinary agent — cystitis only, NOT pyelonephritis.

BOARD TRAP

Fosfomycin treats cystitis, not pyelonephritis or systemic infection. Its niche is single-dose therapy and coverage of resistant/ESBL organisms — not as a tissue agent.

4. Pivmecillinam 400

WHAT YOU SEE

'PIVMEC 400' bottle

WHAT IT ENCODES

Pivmecillinam is a first-line uncomplicated-cystitis agent (widely used in Europe), a beta-lactam with low collateral resistance and a good safety profile in pregnancy. Cystitis-specific; not for pyelonephritis. Recently FDA-approved in the US for uncomplicated UTI.

BOARD TRAP

Pivmecillinam is for uncomplicated cystitis only and should not be relied on for upper-tract or systemic infection — same urinary-agent limitation as nitrofurantoin/fosfomycin.

5. Beta-lactam PO = less effective

WHAT YOU SEE

'BETA-LACTAM PO = LESS EFFECTIVE' shelf banner

WHAT IT ENCODES

Oral beta-lactams (amoxicillin-clavulanate, cephalexin, cefdinir) are SECOND-LINE for cystitis — they have lower efficacy and higher relapse rates than the first-line trio, partly due to inadequate urinary levels and resistance. Reserve them when first-line agents can't be used (e.g., pregnancy combinations).

BOARD TRAP

Don't pick an oral beta-lactam as first choice for routine cystitis — they are less effective with more relapse than nitrofurantoin/TMP-SMX/fosfomycin. They earn their place mainly in pregnancy.

6. FDA AVOID — fluoroquinolone tendon / aorta / nerve

WHAT YOU SEE

Red 'FDA AVOID' triangle: 'TENDON', 'AORTA', 'NERVE' panels, suited man

WHAT IT ENCODES

FDA black-box / safety warnings push fluoroquinolones OUT of first-line use for uncomplicated cystitis because of disabling adverse effects: tendinitis/Achilles RUPTURE (worse with steroids, elderly, transplant), aortic aneurysm/DISSECTION, and peripheral NEUROPATHY (plus QT prolongation, dysglycemia, CNS effects, C. diff). Reserve quinolones for pyelonephritis or when alternatives fail.

BOARD TRAP

Fluoroquinolones are NOT first-line for simple cystitis — the FDA warns to avoid them when alternatives exist due to tendon rupture, aortic dissection, and neuropathy. Their role is complicated UTI/pyelonephritis.

7. Inpatient — IV beta-lactam / aminoglycoside

WHAT YOU SEE

'INPATIENT' arch: 'IV BETA-LACTAM', 'AMINOGLYCOSIDE', knight at bedside

WHAT IT ENCODES

For severe pyelonephritis or sepsis requiring admission, treat IV: a broad-spectrum beta-lactam (ceftriaxone, piperacillin-tazobactam, or a carbapenem for ESBL risk) ± an aminoglycoside. De-escalate to oral once afebrile/improving, guided by culture and sensitivities; total 7–14 days.

BOARD TRAP

Hospitalized pyelonephritis needs tissue-penetrating IV therapy (ceftriaxone/pip-tazo/carbapenem ± aminoglycoside) — urinary-only agents (nitrofurantoin, fosfomycin) are inappropriate for upper-tract/systemic disease.

8. Outpatient pyelonephritis

WHAT YOU SEE

'OUTPATIENT' arch: 'CTX 1 ONCE', 'TMP-SMX 14d', 'CIPRO 7d'

WHAT IT ENCODES

Mild outpatient pyelonephritis: a single IV/IM dose of a long-acting agent (ceftriaxone or an aminoglycoside) followed by oral therapy — a fluoroquinolone (ciprofloxacin 7 days / levofloxacin 5 days) where resistance is low, or TMP-SMX for 14 days if susceptible. Get a urine culture to tailor therapy.

BOARD TRAP

For outpatient pyelonephritis, oral nitrofurantoin/fosfomycin will FAIL (no tissue levels). Choose a fluoroquinolone or TMP-SMX (tissue-penetrating); give an initial long-acting IV dose if quinolone resistance is a concern.

9. Pregnancy — safe agents

WHAT YOU SEE

'PREGNANCY' pregnant figure with green checks: AMP, NITRO, cephalixin

WHAT IT ENCODES

Pregnancy-safe UTI antibiotics: amoxicillin/ampicillin, cephalixin, and fosfomycin; nitrofurantoin is acceptable in the 2nd trimester but AVOID at term (neonatal hemolysis) and in G6PD deficiency. Always treat asymptomatic bacteriuria and cystitis in pregnancy to prevent pyelonephritis/preterm birth.

BOARD TRAP

In pregnancy AVOID fluoroquinolones (cartilage) throughout and TMP-SMX in the 1st trimester (neural tube defects, folate antagonism) and at term (kernicterus). Nitrofurantoin is fine mid-pregnancy but not near delivery.

10. Near term — avoid nitro & sulfa

WHAT YOU SEE

'NEAR TERM' red X over NITRO/TMP, 'KERNIC' sunflower

WHAT IT ENCODES

At/near term, avoid nitrofurantoin (G6PD/neonatal hemolysis) and sulfonamides/TMP-SMX (displace bilirubin → kernicterus in the newborn). Switch to a beta-lactam (amoxicillin, cephalixin) in late pregnancy.

BOARD TRAP

The 'kernicterus' clue means sulfonamide near term — avoid TMP-SMX in late pregnancy; pair that with avoiding nitrofurantoin at term for hemolysis. Beta-lactams are the safe late-pregnancy fallback.

11. Recurrent UTI — ≥6 months / prophylaxis

WHAT YOU SEE

'RECURRENT' calendar 'DAILY ≥6 MONTHS', 'METHENAMINE', 'POST-COITAL'

WHAT IT ENCODES

Recurrent UTI (≥2 in 6 months or ≥3 in a year): options are continuous low-dose prophylaxis (e.g., nitrofurantoin or TMP-SMX) for ≥6 months, POST-COITAL single-dose prophylaxis if intercourse-associated, or self-start therapy. Non-antibiotic measures: methenamine hippurate, topical vaginal estrogen in postmenopausal women, and hydration; cranberry has weak evidence.

BOARD TRAP

For intercourse-related recurrent UTI, post-coital single-dose prophylaxis is preferred over daily dosing. In postmenopausal women, add vaginal estrogen — a frequently overlooked, high-yield non-antibiotic intervention.

12. Vomiting / flank pain → admit

WHAT YOU SEE

'VOMITING', 'FLANK PAIN' figures center

WHAT IT ENCODES

Pyelonephritis with vomiting (can't tolerate orals), hemodynamic instability, pregnancy, severe pain, or failure of outpatient therapy mandates ADMISSION for IV antibiotics and hydration. Inability to keep down oral medication is a key admission trigger even in otherwise stable patients.

BOARD TRAP

Pyelonephritis + persistent vomiting is an admission for IV therapy — sending such a patient home on oral antibiotics they can't keep down is the wrong move.

13. Men / prostate — duration

WHAT YOU SEE

'MEN / PROSTATE' blacksmith: 'FQ / SMX', 'ACUTE 7–14d', 'CHRONIC 4–6w (12w)'

WHAT IT ENCODES

UTI in men is considered COMPLICATED and often involves the prostate. Use tissue- and prostate-penetrating agents: fluoroquinolones or TMP-SMX. Acute bacterial prostatitis: 2–4 weeks (often 7–14 days acute UTI, longer if prostatitis). CHRONIC bacterial prostatitis needs 4–6 weeks (up to 6–12 weeks) because of poor antibiotic penetration into prostatic tissue.

BOARD TRAP

Men's UTI/prostatitis requires prostate-penetrating drugs (fluoroquinolone or TMP-SMX) and LONG courses — nitrofurantoin and fosfomycin penetrate the prostate poorly and are inappropriate for prostatitis.

14. Methenamine — non-antibiotic prophylaxis

WHAT YOU SEE

'METHENAMINE' label in recurrent panel

WHAT IT ENCODES

Methenamine hippurate is a urinary antiseptic: in acidic urine it releases formaldehyde, which is bactericidal without selecting for resistance — useful for recurrent-UTI prophylaxis as an antibiotic-sparing strategy. It needs acidic urine and adequate dwell time and is for prevention, not treatment of active infection.

BOARD TRAP

Methenamine is prophylaxis, not treatment of acute UTI, and is ineffective with urea-splitting organisms (*Proteus*) that alkalinize urine — it works only when urine is kept acidic.